BREEZE

Your Engineering Partner for AI Hardware

BOM optimization | DFM | Prototyping | Manufacturing

3,000+
Component suppliers

12 weeks
Concept to prototype

1,000
Min pilot run

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Building AI Hardware Is Hard.

80% of hardware startups fail at manufacturing, not the idea stage.

Factories won't take your 1K MOQ

Most factories want 10K+ units. We start at 1,000.

Unpredictable NRE costs

You get a clear, fixed NRE quote before any work starts.

IP leaks and design theft

NDA + IP assignment standard on every engagement. Your designs stay yours.

Language and timezone barriers

Native English + Mandarin team. We work your hours.

No DFM feedback until late

We run DFM checks at the design stage, not after you've committed to tooling.

Certification maze

FCC, CE, UL, SRRC -- we handle the paperwork, testing, and timeline.

Concept to Production in 6 Steps

A clear, predictable path from idea to product.

01

Scoping

Share your concept. We assess feasibility, BOM, and ballpark cost.

02

Engineering Design

Schematic, PCB layout, mechanical design, firmware architecture.

03

DFM & Prototyping

Optimize for manufacturability. 3-5 functional prototypes.

04

Tooling & Sourcing

Injection molds, jigs, fixtures. Component sourcing from 3,000+ local suppliers.

05

Pilot Run

50-500 units. Test assembly line, QC process, packaging.

06

Mass Production

Scale to 1K, 10K, 100K+. Ongoing QC, logistics, fulfillment.

Cost Reality Check

$99 AI wearable pendant -- real BOM breakdown @ 5K units

$26.40 / unit avg landed cost

Component	Example	Cost Range
SoC / MCU	ESP32-S3 / nRF5340	$1.80 - $4.50
Memory (Flash + RAM)	4MB + 512KB	$0.60 - $1.20
Microphone Array	MEMS x2	$0.80 - $2.00
Speaker / Haptic	Micro speaker + LRA	$0.50 - $1.50
Battery (LiPo)	250mAh pouch cell	$1.20 - $2.50
BLE + Wi-Fi Antenna	Chip antenna / FPC	$0.30 - $0.80
PCB (4-layer HDI)	30x25mm, ENIG finish	$0.80 - $1.50
Enclosure (injection)	PC/ABS, 2-shot mold	$0.60 - $1.20
Connector + Misc	USB-C, pogo pins, ESD	$0.40 - $0.80
PCBA Assembly	SMT + manual + test	$1.50 - $3.00
Packaging	Box, insert, manual, cable	$0.60 - $1.20
Landed Unit Cost*		$26.40 avg

DFM: Top 10 Pitfalls

Mistakes we've seen (and fixed) across 50+ hardware projects.

1 Thermal runaway

AI inference chips generate 2-5W. Without proper thermal paths, throttling kills performance.

2 Antenna detuning

The human body and enclosure materials shift antenna performance 3-6 dBm. Simulate with real enclosure.

3 Battery certification gap

UN38.3 + IEC 62133 are mandatory for shipping. Budget 6-8 weeks and $3-5K.

4 Underspec'd power supply

Peak current draw during AI inference can be 3x idle. Size your regulator for bursts.

5 Acoustic cavity design

Mic array performance depends on cavity geometry, not just microphone specs.

6 Injection mold draft angles

Below 1.5 degrees on textured surfaces = stuck parts and scratched finish.

7 Test jig as afterthought

Design the test fixture alongside the product. Retrofitting costs 3x and delays 4 weeks.

8 Connector placement

USB-C orientation and position affect waterproofing, drop strength, and user ergonomics.

9 ESD protection gaps

Every external port needs TVS diodes. Skipping them passes lab testing but fails in the field.

10 Silkscreen and markings

FCC ID, CE mark, and recycling symbols need specific sizes and placements per regulation.

Certification Roadmap

What you need, how long it takes, and what it costs.

Certification	Region	Timeline	Budget	Required For
FCC Part 15	United States	4-6 weeks	$3-8K	Any radio transmitter (BLE, Wi-Fi)
CE (RED)	European Union	6-8 weeks	$5-12K	Radio equipment sold in EU/EFTA
UL/IEC 62368	US / Global	8-12 weeks	$8-15K	Audio/video/IT equipment safety
UN38.3	Global	4-6 weeks	$3-5K	Any product with lithium battery
IC (ISED)	Canada	4-6 weeks	$2-4K	Radio devices sold in Canada
SRRC + CCC	China	8-16 weeks	$5-10K	Radio + safety for China market
MIC (Giteki)	Japan	6-10 weeks	$4-8K	Radio equipment sold in Japan

Pro Tip: Start pre-compliance testing during DFM. Saves 4-8 weeks and $5-10K.
We include pre-compliance checks in our standard engineering process.

Free Tools

No signup required. Start estimating your project now.

1 BOM Cost Estimator

Input your specs, get a manufacturing cost ballpark in 60 seconds. Covers SoC, memory, sensors, battery, PCB, enclosure, and assembly.

breezehw.com/tools/bom-estimator

2 DFM Checklist

30-point design-for-manufacturability check covering thermal, mechanical, electrical, and certification pitfalls specific to AI hardware.

breezehw.com/tools/dfm-checklist

3 NRE Quote Simulator

Estimate your total development cost: engineering hours, tooling, prototyping, certification, and project management.

breezehw.com/tools/nre-simulator

4 Certification Navigator

FCC, CE, UL, SRRC, UN38.3 -- which do you need? Get a timeline, cost estimate, and testing lab recommendations.

breezehw.com/tools/cert-navigator

Why Breeze?

Four reasons founders choose us over brokers and trading companies.

Whole Product, Not Just Parts

We build the full device -- electronics, enclosure, firmware, packaging. One team, one responsibility, one timeline.

No Equity, No Surprises

NRE + per-unit pricing. Pure service model. We don't take a piece of your company. Period.

Your IP Stays Yours

NDA + IP assignment is standard on every engagement. You receive full source files, schematics, and manufacturing data.

AI Hardware Native

We understand edge AI thermal management, antenna design for wearables, battery optimization, and low-power ML inference.

Trading companies: 20-40% margin, zero engineering. | In-house: $500K+/yr, 6+ months to hire. | Breeze: ready now.

Ready to Start?

1

Try Our Free Tools

breezehw.com/tools -- no signup

2

Book a Scoping Call

30 min -- feasibility + cost range

3

Get Your NRE Proposal

Fixed pricing, clear timeline

BREEZE

hello@breezehw.com | breezehw.com

No equity. No MOQ surprises. Just engineering.